

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I **AYUSH MOHAPATRA 192411263** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Solution 1:

- Encrypted message: A shift of 4 positions applied to "MEET AT NOON" gives "QIIX EX RSSR".
(M+4=Q, E+4=I, E+4=I, T+4=X, A+4=E, T+4=X, N+4=R, O+4=S, O+4=S, N+4=R).
- Retrieval: An authorized member would apply a reverse shift of 4 positions (or shift forward by 22) to each letter in the ciphertext to retrieve the original message.

Solution 2:

- Original message: "THIS IS A SECRET".
- Justification: The text structure and word lengths suggest the Atbash cipher (where A<->Z, B<->Y, etc.). Reversing the letters in "GSRH RH Z HVXIVG" perfectly decodes to "THIS IS A SECRET". The single letter "Z" decrypting to "A" is a strong indicator of this cipher.

Solution 3:

Step-by-step Transformation:

Message: H E L L O

Keyword: K E Y K E

Encryption process (Message letter + Keyword letter = Ciphertext letter):

$$H(7) + K(10) = R(17)$$

$$E(4) + E(4) = I(8)$$

$$L(11) + Y(24) = J(35 \bmod 26 = 9)$$

$$L(11) + K(10) = V(21)$$

$$O(14) + E(4) = S(18)$$

Encrypted message: "RIJVS".

Influence of repeating keyword: The keyword aligns with the plaintext and shifts each plaintext letter by the position of the corresponding keyword letter. This creates a polyalphabetic substitution, where identical plaintext letters (e.g., "L") are encrypted into different ciphertext letters ("J" and "V") depending on the keyword.

Solution 4:

- a. Detection: Investigators could detect hidden data using visual analysis (e.g., examining LSB planes for anomalies), statistical analysis (e.g., analyzing colour frequencies and structural anomalies like unusual file sizes), and checking the file's metadata for hidden signatures.
- b. Recommendations for strengthening security:
 1. Encrypt the confidential data before embedding it to ensure that even if extracted, it remains unreadable without the decryption key.
 2. Use advanced steganographic techniques, such as randomly distributing the hidden data across the image using a stego-key (instead of sequential embedding), or hiding data in transform domains (like DCT coefficients).

Solution 5:

- a) Original message: "WE ARE DISCOVERED FLEE AT ONCE".(Using a Rail Fence cipher with depth 3, the 25 letters are distributed across the rails as 7, 12, and 6 letters. Reading them out in a zigzag pattern reveals the plaintext).
- b) Vulnerability: Transposition ciphers only rearrange letters without altering their frequencies. Frequency analysis will perfectly match standard English, revealing it as a transposition cipher. A depth of 3 provides very little security against brute-force attacks.
- c) Practical enhancement: Combine the transposition cipher with a substitution cipher (such as Vigenère or AES) to create a product cipher. This introduces both confusion and diffusion, masking letter frequencies and positions.